# Benchmark Evaluation of Reinforcement Learning Agents

LunarLander-v3 Environment Analysis

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## 1 Overview

This benchmark compares three reinforcement learning architectures on `LunarLander-v3`: **Deep Q-Network (DQN)**, **REINFORCE** (policy gradient), and **Advantage Actor-Critic (A2C)**.

To ensure consistency, all agents are evaluated under the same experimental setup:

- Environment: `LunarLander-v3` (discrete action space)
- Seed: `{42}`
- Training duration: 1000 episodes per agent
- Logging format: `episode, reward, length, loss`

**Note:** This report presents results from a single-seed experiment. While it provides useful insights, multi-seed evaluation is recommended for stronger statistical conclusions.

## 2 Final Performance Summary

Final performance is computed as the mean return over the last 100 episodes of training.

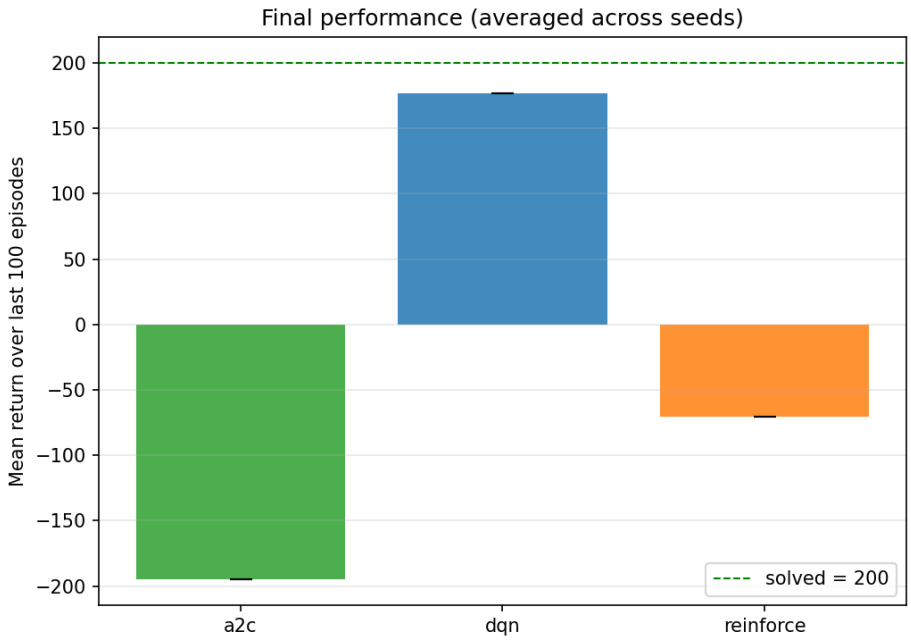


Figure 1: Final mean return for seed 42. The dashed line indicates the commonly used solved threshold (200).

Table 1: Summary of Quantitative Results (seed 42)

| Algorithm | Final Mean Return | Convergence (MA≥200) | Stability (last 100 std) | Interpretation |
|---|---|---|---|---|
| DQN | 176.41 | Episode 850 | 173.82 | Near solved performance |
| REINFORCE | -70.19 | Not reached | 85.51 | Suboptimal convergence |
| A2C | -194.76 | Not reached | 109.30 | Unstable learning dynamics |

# 3 Training Behavior Analysis

## 3.1 DQN Analysis (`curves_dqn.png`)

DQN demonstrates the strongest learning trend among all tested agents.

- **Reward trend:** Gradual improvement with a noticeable increase in later training stages, occasionally surpassing the solved threshold.
- **Learning dynamics:** The TD loss shows fluctuations during mid-training, which is expected as the agent transitions toward more optimal policies.

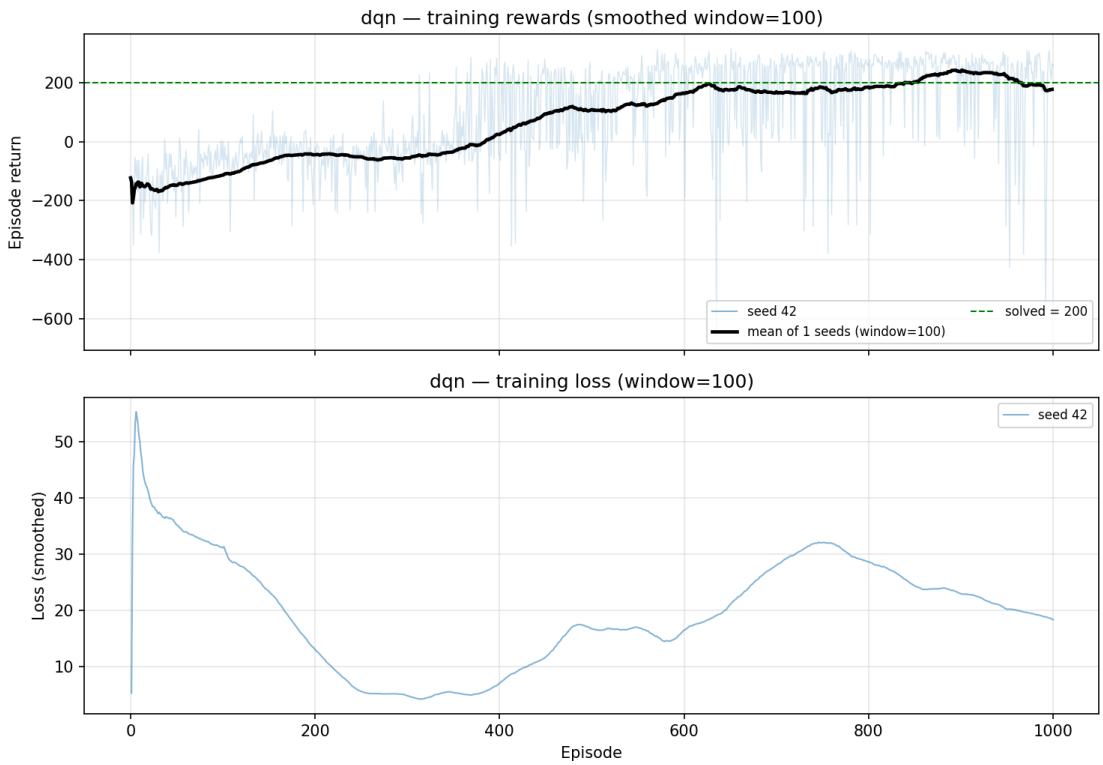


Figure 2: DQN training curve showing steady improvement in performance over time.

## 3.2 REINFORCE Analysis (`curves_reinforce.png`)

REINFORCE shows gradual improvement but remains below the solved threshold within the training budget.

- **Reward trend:** Slow upward movement with high variance typical of Monte Carlo policy gradients.
- **Learning behavior:** Improvement is present but limited by high variance in gradient estimates.

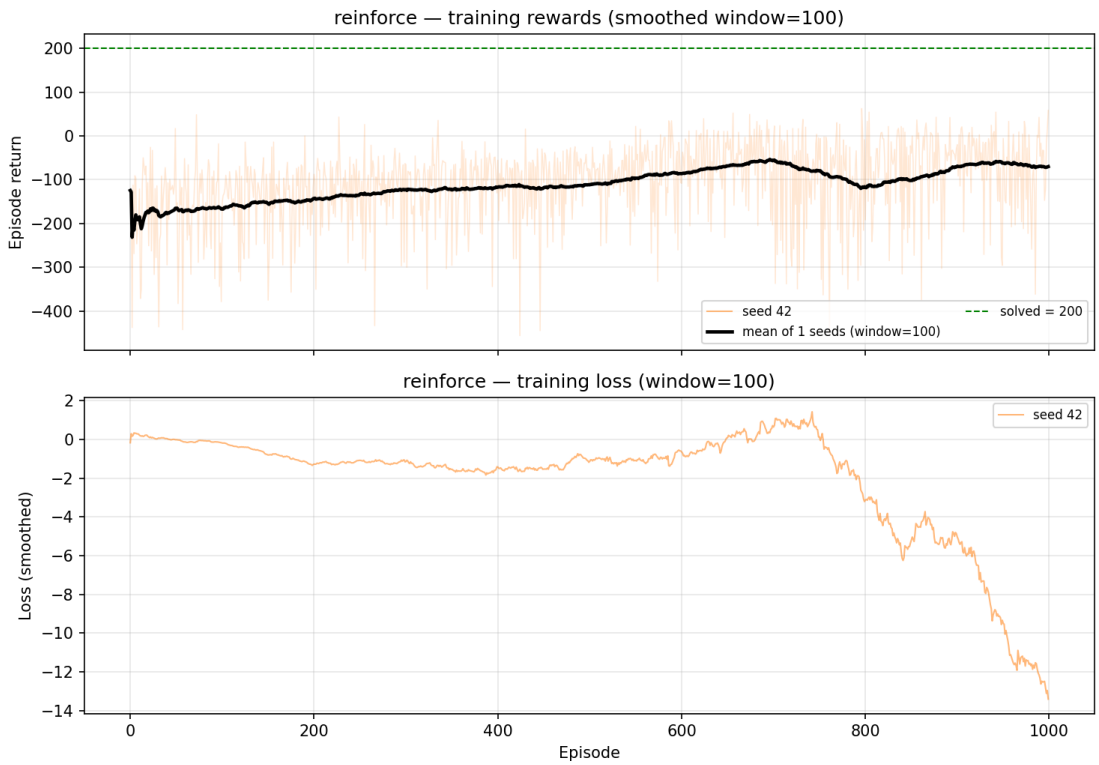


Figure 3: REINFORCE training curve showing gradual but limited improvement.

## 3.3 A2C Analysis (`curves_a2c.png`)

A2C does not show consistent improvement in this experimental setting.

- **Reward trend:** Mostly negative returns throughout training with high variability.
- **Learning dynamics:** Indicates instability in actor-critic coordination under the current hyperparameters.

Figure 4: A2C training curve showing unstable and inconsistent learning behavior.

# 4 Cross-Algorithm Comparison

The following figure compares all algorithms using smoothed reward curves (window size = 100).

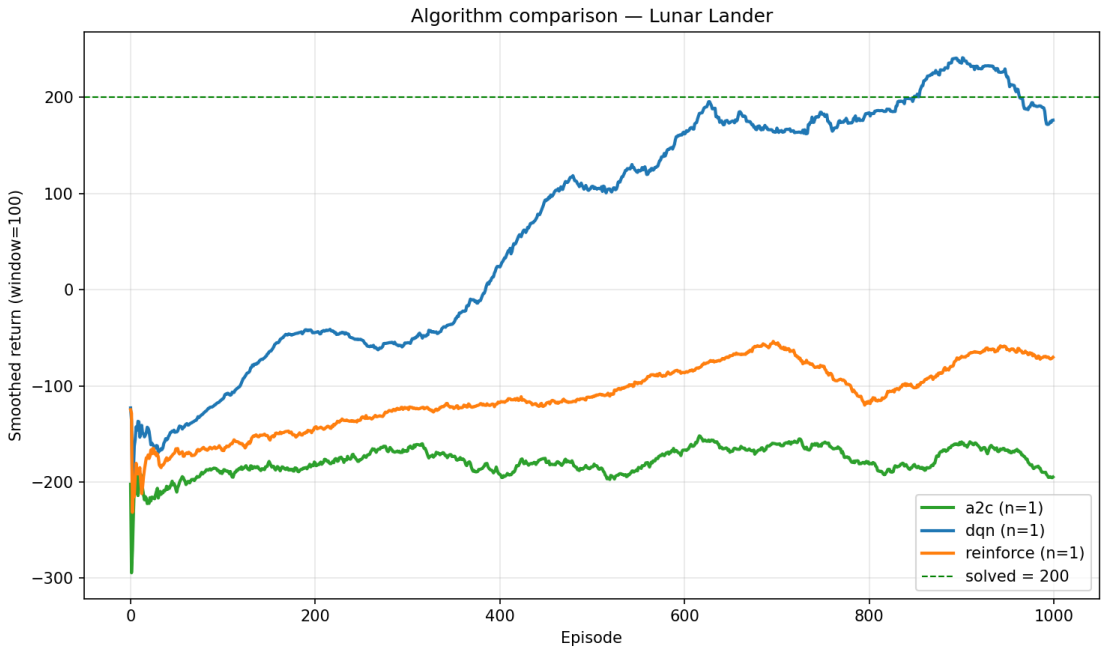


Figure 5: Comparison of smoothed returns across DQN, REINFORCE, and A2C.

DQN shows the strongest learning progression, while REINFORCE and A2C exhibit slower and less stable improvement patterns.

# 5 Conclusion

In this single-seed benchmark (seed = 42), DQN achieves the strongest performance among the evaluated methods, showing the clearest learning progression and highest final return.

However, the final performance remains slightly below the standard solved threshold of 200, indicating that the task is not fully solved under this configuration.

REINFORCE and A2C show learning behavior consistent with their expected limitations under default hyperparameters. Both methods may benefit from additional tuning, longer training, or variance reduction techniques.

Overall, these results highlight the trade-offs between value-based and policy-based reinforcement learning methods in the LunarLander environment.